

Received May 21, 2019, accepted June 20, 2019, date of publication June 26, 2019, date of current version July 12, 2019.

Digital Object Identifier 10.1109/ACCESS.2019.2924933

V-RSIR: An Open Access Web-Based Image Annotation Tool for Remote Sensing Image Retrieval

DONGYANG HOU^{1,2}, ZELANG MIAO¹, (Member, IEEE), HUAQIAO XING³, AND HAO WU⁴

¹School of Geosciences and Info-Physics, Central South University, Changsha 410083, China

²College of Geography and Environment, Shandong Normal University, Jinan 250358, China

³School of Surveying and Geo-Informatics, Shandong Jianzhu University, Jinan 250101, China

⁴National Geomatics Center of China, Beijing 100830, China

Corresponding authors: Huaqiao Xing (xinghuaqiao@126.com) and Hao Wu (wuhao@ngcc.cn)

This work was supported in part by the National Natural Science Foundation of China under Grant 41701443 and Grant 41801308, and in part by the National Key Research and Development Program of China under Grant 2018YFB0505002.

ABSTRACT Benchmark datasets play an important role in evaluating remote sensing image retrieval (RSIR) methods. At present, several small-scale benchmark datasets for RSIR are publicly available on the web and are mostly collected through the Google Map API or other desktop tools. Because the Google Map API requires the users to have programming skills and other collection tools are not publicly available, this may limit the possibility for a wide range of volunteers to participate in generating large-scale benchmark datasets. To address this challenge, we develop an open access web-based tool V-RSIR that allows volunteers to easily participate in generating new benchmark datasets for RSIR. This web-based tool not only facilitates the remote sensing image label and cropping, but also provides image editing, review, quantity statistics, spatial distribution, sharing, and so on. To validate this tool, we recruit 32 volunteers to label and crop remote sensing images by using the tool. Finally, a new benchmark dataset that contains 38 classes with at least 1500 images per class is created. Then, the new dataset is validated by five handcrafted low-level feature methods and four deep learning high-level feature methods. The experimental results show that the handcrafted low-level feature methods perform worse than the deep learning methods, in which the precision at top 5 can achieve 94%. The evaluation results are consistent with our theoretical understanding and experimental results on the PatternNet dataset. This indicates that our web-based tool can help users generating valid benchmark datasets with volunteers for the RSIR.

INDEX TERMS Annotation tool, benchmark dataset, remote sensing image retrieval, volunteers, web-based tool.

I. INTRODUCTION

The recent advances in satellite technology lead to a dramatic growth of remote sensing (RS) images with different resolutions [1], [2]. This has provided new opportunities for various RS applications [3], however, it also results in the significant challenge of retrieving RS images from a considerable volume of RS images [4], [5]. Therefore, it becomes one of the active and emerging research topics to develop efficient remote sensing image retrieval (RSIR) approaches [6], [7].

In general, developing such RSIR approach requires extensive benchmark datasets to evaluate its performance [5], [6].

The associate editor coordinating the review of this manuscript and approving it for publication was Gerardo Di Martino.

The benchmark datasets need to contain a huge amount of RS images labelled with different categories (i.e., airplane, forest and freeway). In the early stages of RSIR, the RSIR benchmark datasets are manually labelled by individuals for personal use and are not made public. These datasets generally cover fewer categories and quantities. For example, a dataset with 60 images and 3 classes is used by Bao and Guo [8], and 3600 images in 6 classes are used by Ferecatu and Boujemaa [9]. Subsequently, with the development of benchmark datasets in similar research fields (e.g., land use/land cover classification and object detection), their published datasets are gradually being used for RSIR [10]–[12]. For instance, the University of California, Merced dataset (UCMD) that is created initially for a land

TABLE 1. Publicly available datasets applied in RSIR.

Dataset	Images per class	Classes	Image number	Resolutions (m)	Size	References
PatternNet	800	38	30400	0.062–4.693	256*	[6]
DLRSD	-	21	2100	0.3	256*	[5]
UCMD	100	21	2100	0.3	256*	[13]
WHU-RS19	50	19	950	up to 0.5	600*	[15]
NWPU-RESISC45	700	45	31500	0.2-30	256*	[3]
RSSCN7	400	7	2800	-	400*	[16]
SIRI-WHU	200	12	2400	2	200*	[17]
AID	220-420	30	10000	0.5-8	600*	[18]

use/land cover classification [13] becomes a most common benchmark dataset for RSIR. Recently, several benchmark datasets specifically for RSIR have been published on the web. For example, Zhou *et al.* [6] presents a novel large-scale RS dataset named 'PatternNet' including 38 classes with 800 images per class, and has evaluated several RSIR methods on 'PatternNet' to establish baseline results; Shao *et al.* [5] presents a multi-labelled dataset termed DLRSD including 17 classes with 2100 images. Table 1 lists several publicly available datasets which have been applied in RSIR. In these datasets, the maximum numbers for per class and all classes are 800 in PatternNet and 31500 in RESISC45, respectively. These datasets play an effective role in RSIR performance evaluation. But they are still small-scale datasets compared to ImageNet, which contains more than 14 million images and covers 20,000 categories [14]. Therefore, new large-scale benchmark datasets for RSIR still need to be generated.

At present, benchmark datasets are often used in RSIR, land use/land cover classification, object detection and its other similar research fields. The methods for generating these benchmark datasets can be divided into 2 categories. One is to label and crop sample images directly from only RS images with the aid of Google Map API, Google Earth, eCognition and other desktop tools. The above datasets are all using this method. However, the Google Map API requires the users to have programming skills, and other desktop tools need to be installed by every participant and require participants to have more remote sensing experience and to conduct several processing steps (e.g., image segmentation and image copy). Besides, the detailed steps of this method are not disclosed in related literature. These shortfalls may limit the possibility for a wide range of volunteers to participate in generating large-scale benchmark datasets. The other is the reference data-based method, which labels images by priori category and location information from reference data. For example, Li *et al.* [19] generated a remote sensing image classification benchmark (RSI-CB) based on points of interests from the OpenStreetMap. Liu and Chen [20] proposed a training sample generation method with the aid of

various globe land cover datasets, such as Globeland30-2010, GlobCover2009, CCILC2010. This method is actually an extension of the first method and it replaces the manual annotation work in the first method with the priori category and location information, which can greatly improve the efficiency of new benchmark datasets generation. But, the prior information of the reference data is not completely correct. For example, an overall classification accuracy of Globeland30 is 80% [21]. Thus, this method still requires a lot of users to verify every image. Moreover, tools for this method are also not publicly available, which is not conducive to the participation of many users. In fact, some works that also require a lot of users have already achieved good results with the help of volunteers through specialized web-based tools [22]–[24]. For example, the Geo-Wiki is a collaborative volunteer-based tool to improve global land cover map validation accuracy [24]. Therefore, developing a specialized web-based tool for RSIR is an effective way to use volunteers to generate large-scale datasets.

In this paper, we describe V-RSIR, an open access web-based tool that allows volunteers to easily participate in generating new benchmark datasets for RSIR. This web-based tool not only facilitates the RS image label and cropping, but also provides image editing, review, quantity statistics, spatial distribution, sharing and so on. In summary, our contributions in this paper are as follows.

1) We first provide a comprehensive review of the approaches for generating benchmark datasets used in RSIR, object detection and other similar fields, by giving a clear categories of these approaches and analyzing their advantages and disadvantages.

2) We introduce crowdsourcing ideas into RS image annotation for RSIR, and propose and implement an open access web-based image annotation tool (V-RSIR) for RSIR. Compared with other tools, the tool allows multiple volunteers around the globe to simultaneously label RS images without installing any plug-ins (except for web browsers).

3) We construct a new RSIR benchmark dataset using the above tool V-RSIR. Unlike other datasets that are only distributed in several major cities in a country, the new dataset is distributed in most countries around the world. This leads to better intra-class diversity of the dataset, which puts higher requirements on the RSIR approaches. Our subsequent comparative experiments illustrate this point. Besides, the dataset is publicly available through our tool V-RSIR.

The remainder of this paper is organized as follows. Section II overviews the requirement analysis, design and implementation of the V-RSIR tool. Section III describes a work-through example including results and analysis. Section IV provides some discussions followed by the conclusion and future work in Section V.

II. DESIGN AND IMPLEMENTATION OF THE TOOL

A. REQUIREMENT ANALYSIS

Based on the analysis of the existing image annotation tools and the production process of the current benchmark datasets,

we give a brief summary of the main requirements of a new image annotation tool.

1) VOLUNTEER PARTICIPATION

Currently, benchmark dataset production for RSIR is a time-consuming and labor-intensive job. However, the current image annotation tools for RSIR work independently by a few users without collaborating with volunteers on the web. Besides, volunteers have been successfully used in labor-intensive jobs, such as OpenStreetMap for creating map [23] data and Geo-Wiki [24] for improving global land cover. Therefore, the new tool should allow volunteers to easily participate in image collecting.

2) CATEGORY LABELLING AND IMAGE CROPPING

According to the needs of RSIR methods evaluation, each image in benchmark datasets should have a correct category and fixed size. Therefore, the new tool should support category labelling and image cropping with a fixed size.

3) EASY CATEGORY SEARCH AND IDENTIFICATION

Generally, volunteer may have less remote sensing experience to identify object categories. Besides, it is difficult and time consuming to find an object in a certain category in a global scale. Therefore, the new tool should provide the ability to speed up the search and identification of object categories.

4) HIGH QUALITY LABELLING

Volunteer participation can alleviate labor-intensive annotations, but it easily leads to quality problems due to mislabelling by volunteers. Therefore, the new tool should have the ability to ensure high quality labelling.

5) SPATIAL DISTRIBUTION OF BENCHMARK DATASETS

The currently publicly available datasets only describe spatial distribution of images in textual form. For example, users only know that the PatternNet is from RS images in the United States cities, and they do not know the specific spatial location of each images [6]. In fact, spatial distribution of a dataset can help users to fully understand the dataset and to create a new dataset. Therefore, the new tool should allow users to view the spatial distribution of datasets.

6) OPEN AND DYNAMIC

Many benchmark datasets are still small-scale and their numbers are constant. This may result in these datasets not being suitable for more complex RSIR models in the future, because a more complex model indicates more image training samples [19]. Hence, the new tool for RSIR should be designed to allow generated datasets to grow over time.

B. V-RSIR ARCHITECTURE

The fundamental design principle is to leverage distributed online image map resources to support RSIR benchmark

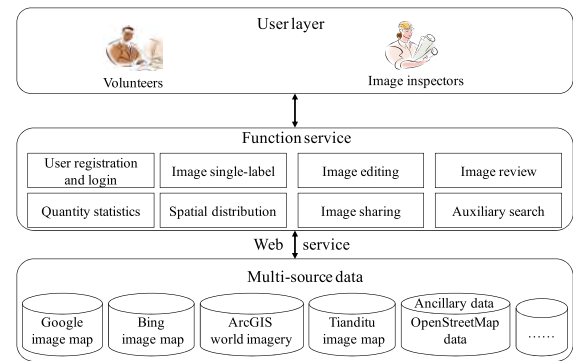


FIGURE 1. Conceptual architecture of the V-RSIR tool.

dataset production by volunteers. The conceptual architecture of V-RSIR, shown in Fig. 1, involves the following three layers.

The multi-source data layer lies at the bottom of the V-RSIR architecture and aims to integrate different online image map resources and other ancillary data. Online image map resources are served as primary RS image source data for RSIR benchmark dataset. In order to generate diversity datasets, 4 different online image resources with different spatial resolutions, different phases and different sensors are integrated into the V-RSIR tool through their service addresses. These resources are Google image map, Bing image map, ArcGIS word imagery and Tianditu image map. Ancillary data (e.g., OpenStreetMap data) could be used to improve the efficiency of object search and identification.

The function service is the core component of the V-RSIR. Its goal is to generate, manage and visualize RS image datasets. It includes user registration, login, image single-label, editing, review, quantity statistics, spatial distribution, sharing, auxiliary search and so on. Image single-label provides a means that participants can contribute their efforts to label image category and capture image screenshot from the above online image map resources. Image editing supports participants to delete their own annotated images which have wrong category information. Image review allows RS professionals to check whether category information of the labelled images is correct and delete the incorrect images. These two functions of image editing and review can ensure the quality of the labelled images to a certain extent. Quantity statistics and spatial distribution are helpful to understand the overall situation of the datasets and the spatial details of each image, respectively. Image sharing provides sharing of the checked datasets via the web. Auxiliary search aims to help users quickly locate, find and identify targets that should be labelled.

The user layer includes volunteers and image inspectors. The volunteers can label, crop and edit RS images. Besides, they also can browse the quantity statistics and spatial distribution of labelled images and download the checked images. In addition to the above permissions, the image inspectors are mainly responsible for checking whether the image is labelled correctly by volunteers.

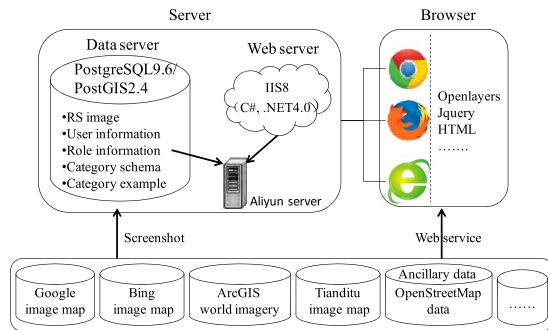


FIGURE 2. System implementation architecture.

C. SYSTEM IMPLEMENTATION FRAMEWORK

The V-RSIR tool has been implemented according to the requirements analysis and architectural design discussed in the previous section. Its implementation adopts a browser/server (B/S) architecture, as shown in Fig. 2.

The sever includes data server and web server. The data server is established using an open source object-relational database PostgreSQL9.6 (<https://www.postgresql.org>) and its spatial extension PostGIS2.4 (<http://www.postgis.net/2017/09/30/postgis-2.4.0/>).

PostgreSQL with PostGIS provides numerous spatial types and functions for performing sophisticated spatial data storage, queries and operations. This can help the V-RSIR tool to store the spatial bounding box of a cropped image and achieve overlapping area reminders.

In the V-RSIR tool, the data sever stores RS image data, user information, role information, category schema, as well as category examples. RS image data contains many RS screenshot of the above online image map resources and related information, such as category, spatial bounding box, source, spatial resolution, size, zoom level, image labeller, image checker and so on. User information stores user id, username, encrypted password, mailbox, country, etc. Role information stores permission actions for volunteers and reviewers. Category schema stores the classification system information, such as Chinese name and English name of the category. Category examples store sample images for each category, which can help volunteers to identify image categories.

The web server is built based on IIS8 (Microsoft Internet Information Server 8) and implemented by using the C# programming language and the Microsoft NET Framework 4.0. The PostgreSQL/PostGIS data server and the web server instances run on an Aliyun server, containing one 2-core 2.50 GHz Intel(R) Xeon(R) Platinum 8163 CPU, 4 GB memory, 200GB disks and an Internet connection with a bandwidth of 5 MB/s. The Aliyun server runs windows server 2012 R2 operating system.

The browser side is implemented using HTML, JQuery, an open source JavaScript library OpenLayers and its geocoder extension ol-geocoder (<https://github.com/jonataswalker/ol-geocoder>). In the V-RSIR tool, the

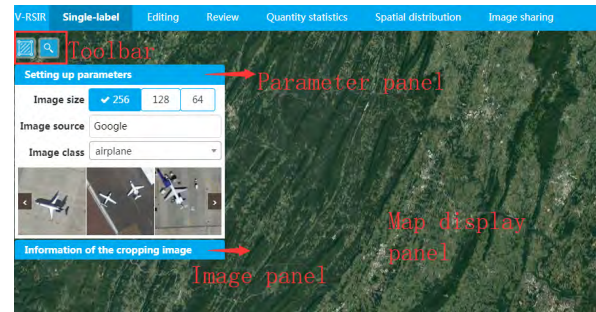


FIGURE 3. The main interface of image single-label.

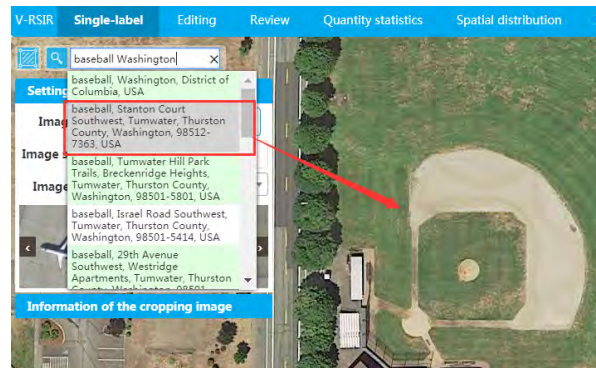


FIGURE 4. The interface of auxiliary search about the example "baseball Washington".

OpenLayers library is used to integrate, visualize and manage the online image map resources through their service addresses. The extension ol-geocoder provides Nominatim search API, which is a tool to search OpenStreetMap data by name and address.

D. MAIN FUNCTIONS OF THE V-RSIR TOOL

The V-RSIR tool is now available at <http://www.geoinfobar.com:5321>, which contains three demonstration videos to help users use the proposed tool.

1) IMAGE SINGLE-LABEL

Volunteers or image inspectors who log in the tool will be presented with the interface of image single-label, as shown in Fig. 3. The interface is divided into five separate components: map display panel, toolbar, parameter panel, image panel and prompt panel, which is hidden by default. The map display panel is responsible for displaying image maps.

The toolbar contains two small functions. One is the function of auxiliary search with the icon of magnifier. It utilizes priori information (e.g. category and position) of auxiliary data to easily find and identify image category. For example, if users want to label baseball field category, they can enter the keywords of "baseball Washington" into the search box, some similar information will be displayed in the drop-down box (Fig. 4).

The other is the function of image selection with the icon of square. It is used to draw a yellow bounding box to limit the spatial scope of the image to be labelled (Fig. 5).

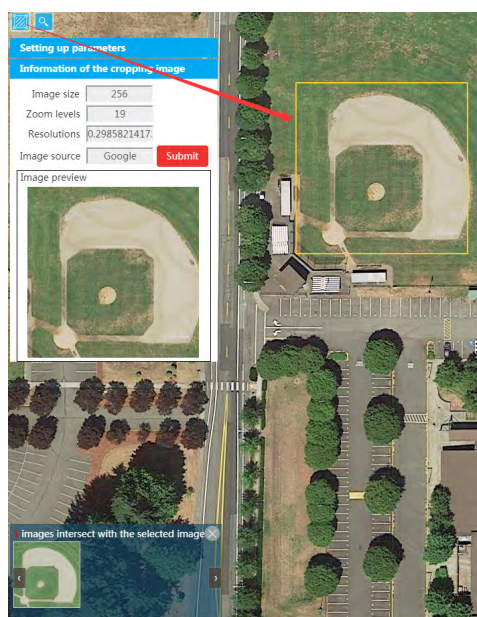


FIGURE 5. The Interface of parameter panel and prompt panel.

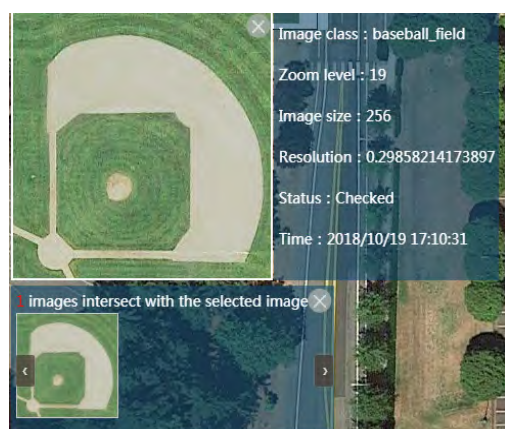


FIGURE 6. The interface of details of labelled images.

The parameter panel allows volunteers to select image size, image source, image category. Moreover, it can display some sample images of the selected category. The image panel is responsible for displaying details (e.g. size and source) about the image to be cropped (Fig. 5). The button of 'Submit' on the image panel can crop the final image and store it into the database. The prompt panel will appear in the bottom right corner of the page (Fig. 5), if other volunteers have already marked the image around this bounding box. The prompt panel allows volunteer to browse the details of the labelled images (Fig. 6).

2) IMAGE EDITING

Fig. 7 presents the interface of image editing. In this module, volunteers can browse their own labelled images by category. Inaccurate images can be deleted individually or in batches through 'delete' button or 'bulk delete' button.

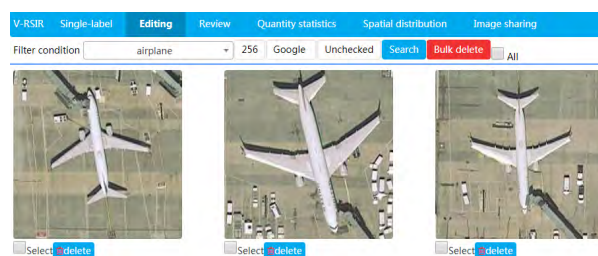


FIGURE 7. The interface of image editing.

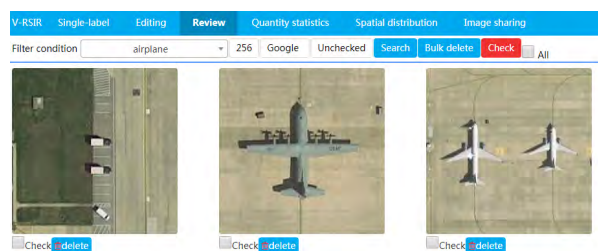


FIGURE 8. The interface of image review.



FIGURE 9. The interface of quantity statistics.

3) IMAGE REVIEW

Occasionally, volunteers may make mistakes due to their negligence and not all volunteers follow the instructions to label images. The solution to address these issues is having professionals to review the images again. Therefore, the V-RSIR tool provides the function of image review (Fig. 8). The image inspectors can delete incorrect images individually or in batches through 'delete' button or 'bulk delete' button. Besides, the correct images should be submitted to the benchmark dataset by clicking 'Check' button.

4) QUANTITY STATISTICS

Users can browse the number of checked and unchecked images through the 'Statistics' button (Fig. 9). The result can be displayed in the form of histogram or line chart by using the library of ECharts (<https://echarts.baidu.com/>).

5) SPATIAL DISTRIBUTION OF IMAGES

Users can browse the spatial distribution of checked and unchecked images through the 'Spatial' button (Fig. 10). The result is displayed in the form of clusters. Clicking one circle

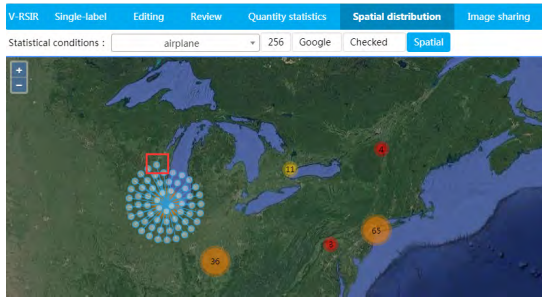


FIGURE 10. The interface of spatial distribution.

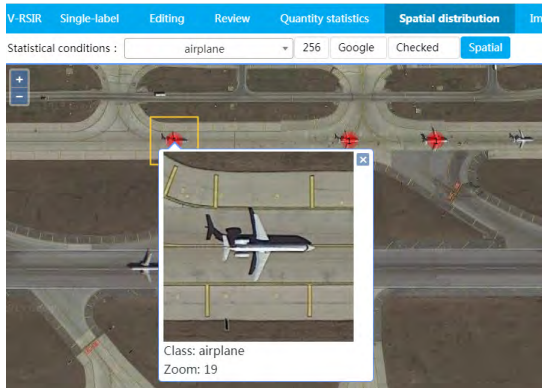


FIGURE 11. The interface of image details. The details are obtained by click the circle of the red box in Figure 10.

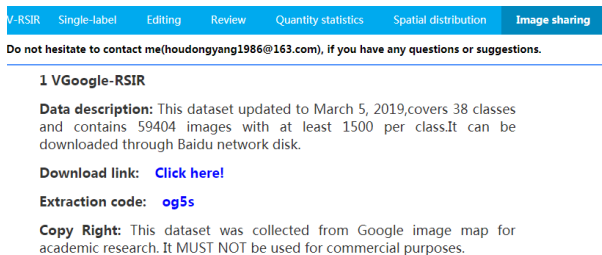


FIGURE 12. The interface of image sharing.

of the clusters, the map will locate to the position of the corresponding image and show image details (Fig. 11).

6) IMAGE SHARING

At present, the labelled data can be acquired by the download link of Baidu cloud disk (Fig. 12).

III. WALK-THROUGH EXAMPLE AND RESULTS

In this section, a walk-through example is presented to demonstrate the usefulness of the V-RSIR tool.

A. WORKFLOW OF THE WALK-THROUGH EXAMPLE

The walk-through example is to construct a new benchmark dataset from the Google image map using the V-RSIR tool. The new dataset is termed as VGoogle-RSIR and adopts the classification scheme of PatternNet dataset, which covers 38 classes, such as airplane, baseball field, basketball court and so on [6]. Besides, the example requires the labelled

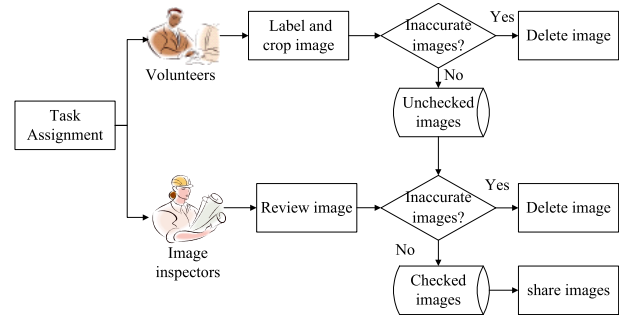


FIGURE 13. The workflow of the walk-through example.

images with the size of 256*256 pixels. With the advantage of working in a university, 32 student volunteers with remote sensing background are recruited to participate in this walk-through example, and 6 of them are selected as image inspectors. In order to shorten the experiment time, the 6 inspectors are also involved in labelling and cropping images, but they will cross review each other's images.

Fig. 13 illustrates the workflow of the walk-through example, which describes different roles and their operations with system functions. First, each volunteer is assigned with 1 or 2 categories annotation work and each image inspector is assigned with 6 or 7 categories of inspection work. Then, the volunteers will label and crop images by using the function of image single-label (Fig. 3 to 6). In this process, each volunteer also needs to use the editing function to check whether they have the inaccurate images. If there are inaccurate images, they need to perform the delete operation (Fig. 7). At the same time, the image inspectors need to check the image labelled by the volunteers one by one using the function of image review (Fig. 8). They should delete inaccurate images and submit correct images into the database. Finally, the checked images will be shared via the V-RSIR tool (Fig. 12).

B. RESULTS OF THE WALK-THROUGH EXAMPLE

In a two-week period, the 32 volunteers spend an average of 30 minutes to an hour a day to complete this walk-through example. Finally, the new VGoogle-RSIR benchmark dataset covers 38 classes and contains 59,404 images with at least 1500 per class. It has spatial resolutions ranging from 0.075 m to 9.555 m and takes up 8.70 GB of storage. The format of the image is TIFF (Tagged Image File Format), such as "airplane000bd572-cb8c-2a44-13a7-7c671708fb91.tiff" and "basketball_court00edb0eb-10d0-768b-42f2-61cea9021918.tiff".

Fig. 14 represents some example images from the VGoogle-RSIR benchmark dataset. Fig. 15 represents the spatial distribution of the VGoogle-RSIR dataset through the function of "spatial distribution". The images in this dataset are mainly distributed in North America, Europe and South America.

Fig. 16 depicts the evolution of the VGoogle-RSIR dataset since the web-based tool went online. The histogram reflects the number of images labelled each day. The Line chart

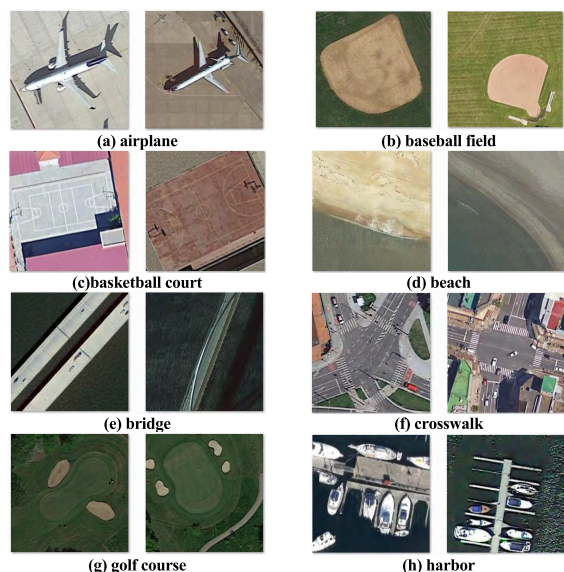


FIGURE 14. Some example images from the VGoogle-RSIR dataset.



FIGURE 15. Spatial distribution of the VGoogle-RSIR dataset.

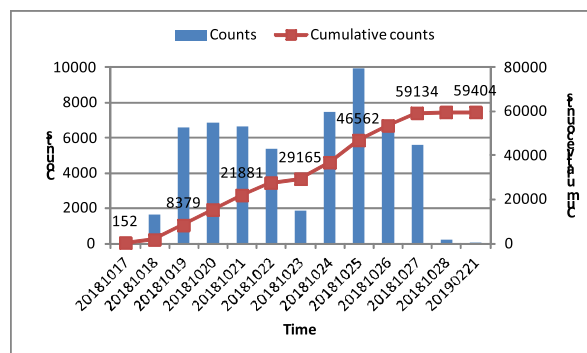


FIGURE 16. Evolution of the VGoogle-RSIR dataset over time.

reflects the cumulative number of images labelled. This shows that the number of RS images increased steadily and indicates that the V-RSIR tool allows generated RS datasets to grow over time.

C. RESULT VERIFICATION

Five handcrafted low-level feature methods and four deep learning high-level feature methods are selected to verify the dataset's ability of evaluating different RSIR methods.

The nine methods are HSV color (hue, saturation, value), Gabor texture features, HOG (histogram of oriented gradient), LBP (local binary pattern), BOW (bag of visual words), VGG-16 (Visual Geometry Group), VGG-19, ResNet50 and DenseNet (dense convolutional network). Six scales and eight orientations are set in the Gabor method. A dictionary size of 64 is set in the BOW method. The deep features are extracted from the first fully-connected layer (f1 feature) and the second fully-connected layer (f2 feature) in the VGG-16 and VGG-19 methods. During the deep learning training process, epoch number is set to 20 as described in [25]. 50 images from each class of the VGoogle-RSIR dataset are randomly selected as the test set (named as google_test) for retrieval performance evaluation and the remaining images are used as the training set (named as google_train). Besides, 50 images from each class of the training set are also randomly selected for validating the deep learning models. In other words, there are 1900 images in the test set and 57504 images in the training set.

Three performance metrics are adopted to evaluate the retrieval performance. The first metric is average normalized modified retrieval rank (ANMRR), for which lower values mean better performance. The remaining two are mean average precision (mAP) and precision at k (Pk where k is the number of retrieved images), for which higher values mean better performance [6], [25]. In the experiment, the three metrics' values are the averages of all queries.

Table 2 shows the performance of these methods. The handcrafted low-level feature methods perform worse than the four deep learning high-level feature methods in the VGoogle-RSIR dataset. Besides, the performance of the DenseNet method is higher than the performance of other methods. The order of performance is: DenseNet > ResNet50 > VGG19_f2 > VGG16_f2 > VGG16_f1 > VGG19_f1 > LBP (BOW) > Gabor(HSV) > HOG. The comparison results are consistent with our theoretical understanding that deep learning methods outperform low-level feature methods. This proves that the dataset labelled by the V-RSIR tool can effectively evaluate the retrieval performance.

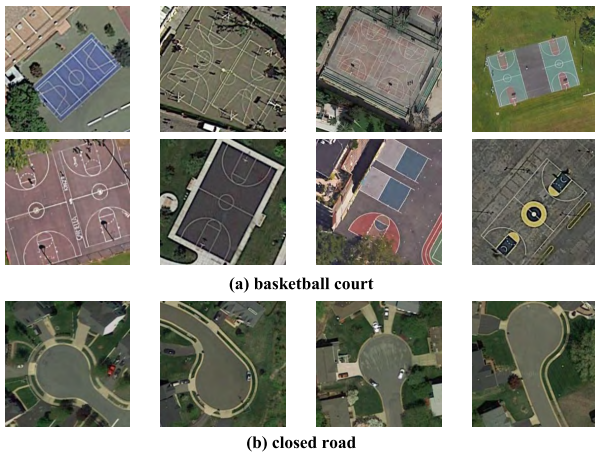
Table 3 depicts the results of the DenseNet method for each class. It shows that the retrieval performances of different categories are different. The category of the basketball court with the number 2 has the lowest accuracy and the category of the closed road with the number 8 has the highest accuracy. The reason may be that the style of the basketball court is varied and the style of the closed road is single in the VGoogle-RSIR datasets (Fig. 17). See Appendix A for the category code table.

D. RESULTS OF COMPARATIVE EXPERIMENTS

To further verify the reliability of the above experimental results, we also conduct some experiments on the PatternNet dataset using the same nine methods. Experimental parameters and dataset partitioning are the same as the above experiments. In other words, 50 images from each class of the PatternNet dataset are randomly selected as the test set

TABLE 2. The results on the VGoogle-RSIR dataset.

Method	ANMRR	mAP	P5	P10	P20	P50	P100	P1000
HSV	0.81	0.12	0.55	0.50	0.45	0.37	0.32	0.17
Gabor	0.80	0.12	0.45	0.42	0.38	0.33	0.29	0.17
LBP	0.78	0.13	0.61	0.56	0.51	0.44	0.38	0.19
HOG	0.89	0.09	0.36	0.32	0.29	0.24	0.21	0.10
BOW	0.74	0.18	0.53	0.50	0.47	0.43	0.39	0.24
VGG16_f1	0.67	0.26	0.81	0.78	0.74	0.68	0.63	0.34
VGG16_f2	0.55	0.38	0.88	0.86	0.84	0.80	0.75	0.47
VGG19_f1	0.71	0.24	0.77	0.74	0.70	0.64	0.58	0.30
VGG19_f2	0.48	0.45	0.90	0.88	0.87	0.83	0.80	0.55
ResNet50	0.41	0.50	0.88	0.86	0.85	0.82	0.79	0.58
DenseNet	0.29	0.63	0.94	0.93	0.92	0.90	0.88	0.71

**FIGURE 17.** Some examples of basketball court and closed road.

(named as pn_test) and the remaining images are used as the training set (named as pn_train).

Table 4 shows the results on the PatternNet dataset. It is apparent from this table that the order of performance is also: DenseNet > ResNet50 > VGG19_f2 > VGG16_f2 > VGG16_f1 > VGG19_f1 > LBP (BOW) > Gabor(HSV) > HOG. The finding is consistent with the results on our VGoogle-RSIR dataset, which indicates that our VGoogle-RSIR dataset and our V-RSIR tool are effective and practical.

Besides, it is interesting to note that each retrieval method performs better on the PatternNet dataset than our VGoogle-RSIR dataset. This does not mean that our dataset is not as good as the PatternNet dataset. In order to prove this statement, we have done some cross-over experiments. In the cross-over experiments, two sets of comparison experiments are set up. The first (named as pn_google) is to use the pn_train as the training set and the google_test as the test set. The second (named as google_pn) is to use our google_train as the training set and pn_test as the test set. Table 5 and Table 6 show the results of the pn_google and google_pn, respectively.

It can be seen from the data in Table 5 and Table 6 that the performance of each retrieval method in google_pn is slightly

TABLE 3. The results of THE DenseNet METHOD FOR Each CLASS.

Class	ANMRR	mAP	P5	P10	P20	P50	P100	P1000
0	0.15	0.81	1.00	0.99	0.99	0.99	0.98	0.88
1	0.27	0.66	0.94	0.94	0.93	0.91	0.89	0.73
2	0.51	0.37	0.78	0.77	0.73	0.68	0.65	0.45
3	0.33	0.56	1.00	1.00	0.98	0.95	0.92	0.63
4	0.45	0.45	0.88	0.86	0.84	0.80	0.78	0.53
5	0.52	0.38	0.91	0.89	0.87	0.81	0.77	0.47
6	0.27	0.66	1.00	0.99	0.99	0.98	0.97	0.78
7	0.39	0.53	1.00	0.99	0.99	0.97	0.96	0.70
8	0.02	0.97	1.00	1.00	1.00	1.00	0.99	0.98
9	0.27	0.66	0.89	0.89	0.90	0.88	0.87	0.73
10	0.27	0.66	0.94	0.92	0.91	0.89	0.88	0.73
11	0.28	0.63	0.91	0.91	0.92	0.91	0.89	0.69
12	0.30	0.61	0.93	0.92	0.91	0.88	0.87	0.68
13	0.27	0.65	0.89	0.87	0.87	0.86	0.85	0.72
14	0.11	0.85	1.00	1.00	1.00	0.99	0.99	0.91
15	0.41	0.49	0.94	0.94	0.93	0.91	0.88	0.57
16	0.10	0.86	0.99	0.99	0.98	0.98	0.98	0.91
17	0.13	0.82	0.98	0.97	0.97	0.97	0.97	0.88
18	0.18	0.75	0.98	0.98	0.97	0.96	0.95	0.80
19	0.15	0.80	1.00	0.99	0.99	0.97	0.97	0.86
20	0.48	0.39	0.74	0.71	0.70	0.65	0.63	0.45
21	0.39	0.53	0.94	0.93	0.90	0.88	0.85	0.61
22	0.08	0.89	0.99	0.99	0.99	0.98	0.98	0.93
23	0.24	0.68	0.95	0.93	0.92	0.91	0.90	0.73
24	0.12	0.85	1.00	0.99	0.99	0.99	0.98	0.90
25	0.36	0.57	0.90	0.88	0.87	0.85	0.83	0.66
26	0.38	0.53	0.94	0.92	0.91	0.88	0.85	0.61
27	0.22	0.71	0.98	0.98	0.98	0.97	0.96	0.81
28	0.39	0.50	0.95	0.92	0.90	0.87	0.83	0.58
29	0.19	0.74	0.99	0.99	0.99	0.97	0.95	0.80
30	0.34	0.57	0.99	0.99	0.97	0.95	0.93	0.68
31	0.47	0.45	0.95	0.95	0.94	0.92	0.90	0.60
32	0.36	0.54	0.83	0.84	0.81	0.80	0.78	0.61
33	0.42	0.49	0.81	0.79	0.77	0.75	0.73	0.56
34	0.13	0.83	1.00	1.00	1.00	1.00	1.00	0.90
35	0.44	0.46	0.89	0.89	0.87	0.85	0.83	0.59
36	0.41	0.49	0.91	0.86	0.84	0.80	0.77	0.56
37	0.30	0.62	0.95	0.94	0.92	0.90	0.88	0.72

better than that of pn_google. The DenseNet method has the biggest performance gap between the two experiments. A possible explanation for the results might be that our VGoogle-RSIR dataset is extracted from a global scale and has better intra-class diversity, when the PatternNet dataset is only extracted from US cities and has lower intra-class diversity.

TABLE 4. The results on the patternnet dataset.

Method	ANMRR	mAP	P5	P10	P20	P50	P100	P1000
HSV	0.69	0.24	0.68	0.63	0.58	0.51	0.46	0.22
Gabor	0.66	0.26	0.68	0.64	0.60	0.54	0.48	0.24
LBP	0.66	0.27	0.73	0.69	0.64	0.57	0.51	0.25
HOG	0.87	0.12	0.41	0.37	0.32	0.26	0.22	0.09
BOW	0.65	0.27	0.68	0.64	0.61	0.55	0.50	0.25
VGG16_f1	0.64	0.31	0.84	0.81	0.77	0.70	0.62	0.25
VGG16_f2	0.50	0.44	0.91	0.89	0.87	0.82	0.77	0.35
VGG19_f1	0.66	0.30	0.80	0.77	0.73	0.66	0.59	0.23
VGG19_f2	0.45	0.50	0.92	0.90	0.88	0.84	0.80	0.39
ResNet50	0.29	0.65	0.93	0.92	0.90	0.88	0.85	0.52
DenseNet	0.18	0.78	0.97	0.97	0.96	0.94	0.93	0.60

TABLE 5. The results of the pn_google.

Method	ANMRR	mAP	P5	P10	P20	P50	P100	P1000
HSV	0.88	0.10	0.17	0.16	0.15	0.15	0.14	0.09
Gabor	0.89	0.09	0.12	0.12	0.12	0.12	0.11	0.08
LBP	0.90	0.08	0.12	0.12	0.12	0.11	0.11	0.08
HOG	0.92	0.08	0.17	0.16	0.15	0.13	0.11	0.06
BOW	0.78	0.17	0.36	0.34	0.33	0.30	0.28	0.16
VGG16_f1	0.71	0.24	0.58	0.55	0.53	0.48	0.44	0.21
VGG16_f2	0.62	0.33	0.66	0.64	0.62	0.58	0.54	0.27
VGG19_f1	0.74	0.23	0.56	0.53	0.50	0.46	0.41	0.18
VGG19_f2	0.59	0.36	0.67	0.65	0.63	0.60	0.56	0.29
ResNet50	0.66	0.28	0.40	0.39	0.39	0.38	0.37	0.25
DenseNet	0.62	0.33	0.44	0.43	0.43	0.42	0.41	0.28

TABLE 6. The results of the google_PN.

Method	ANMRR	mAP	P5	P10	P20	P50	P100	P1000
HSV	0.87	0.08	0.24	0.23	0.21	0.19	0.17	0.11
Gabor	0.87	0.08	0.18	0.17	0.17	0.16	0.15	0.11
LBP	0.89	0.07	0.15	0.15	0.14	0.13	0.12	0.09
HOG	0.91	0.07	0.22	0.20	0.19	0.16	0.14	0.08
BOW	0.76	0.15	0.44	0.41	0.40	0.36	0.34	0.21
VGG16_f1	0.74	0.20	0.63	0.60	0.58	0.53	0.48	0.26
VGG16_f2	0.65	0.29	0.73	0.71	0.69	0.65	0.61	0.36
VGG19_f1	0.77	0.19	0.61	0.59	0.56	0.51	0.46	0.23
VGG19_f2	0.58	0.36	0.77	0.76	0.74	0.71	0.67	0.44
ResNet50	0.50	0.40	0.67	0.66	0.65	0.64	0.62	0.47
DenseNet	0.42	0.49	0.76	0.75	0.74	0.73	0.71	0.56

IV. DISCUSSION

Our walk-through example confirms that the V-RSIR tool can help users to generate new benchmark datasets for RSIR. Compared with the current tools, the V-RSIR tool is open access and available via the web, which provides a

user-friendly interface without installing any desktop or professional plugins. Furthermore, the V-RSIR tool gives access to multi-source online RS image maps to allow many volunteers participating in image annotation at the same time, which provides the possibility to generate large-scale datasets like ImageNet.

Despite the advantages of the V-RSIR tool, there are still several aspects that can be done to improve usability and efficiency of the tool.

A. EXTENSION OF CLASSIFICATION SCHEMES

Currently, the V-RSIR tool only integrates the classification scheme of PatternNet dataset. However, there are many different classification schemes in the field of remote sensing applications, such as scheme of GlobeLand30 [21] and WHU-RS19 [15]. In order to meet different category retrieval needs, the V-RSIR tool should add multiple classification schemes or allow users to define categories.

B. EXTENSION OF NEW RS IMAGE RESOURCES AND ORIGINAL RS IMAGES

With the satellite technology developed, multiple RS images, especially high-resolution images, are increasingly becoming available [2], [26]. The V-RSIR tool will have to be able to ensure access to new RS image resources. Allowing users to add new image resources through the web map service is one way to solve this problem. At present, the V-RSIR tool crops RS images using screen capture technology, which results in only obtaining RGB or RGBA images. In fact, original RS images have multi-band (multi-spectral) information, which is useful for improving retrieval accuracy. Therefore, the V-RSIR tool needs to have a large amount of original RS images. However, it is unrealistic for us to collect and store so many original RS images. Fortunately, the Google Earth Engine (GEE) provides a multi-petabyte catalog of satellite images (e.g., Landsat, MODIS, Sentinel satellites etc.) and supports a dominant cloud-based geospatial processing [27], [28], which could facilitate the V-RSIR tool to generate new datasets with original RS images.

C. EXTENSION OF MULTI-LABEL AND AUTOMATION

As described in [5] and [29], a RS image might have multiple categories and less efforts have been contributed to generate multi-label datasets for RSIR. Therefore, developing multi-label function is an improvement direction of the V-RSIR tool. For multi-label RS images, the boundaries of the categories or their regions of interests (ROI) are irregular, which makes it difficult and time-consuming for volunteers to accurately sketch. Therefore, some segmentation algorithms (e.g. fully convolutional networks [30] and mask R-CNN [31]) or ROI -related methods (e.g. human-centric brain imaging space [32], [33]) should be considered in the development of multi-label functions.

Moreover, another improvement direction is to improve the automation of annotation. As the annotated RS images grow, various object detection algorithms (e.g., a rotation-

TABLE 7. The category code table.

Class	Code	Class	Code
airplane	0	mobile home park	19
baseball field	1	nursing home	20
basketball court	2	oil gas field	21
beach	3	oil well	22
bridge	4	overpass	23
cemetery	5	parking lot	24
chaparral	6	parking space	25
christmas tree farm	7	railway	26
closed road	8	river	27
coastal mansion	9	runway	28
crosswalk	10	runway marking	29
dense residential	11	shipping yard	30
ferry terminal	12	solar panel	31
football field	13	sparse residential	32
forest	14	storage tank	33
freeway	15	swimming pool	34
golf course	16	tennis court	35
harbor	17	transformer station	36
intersection	18	wastewater treatment plant	37

invariant convolutional neural network model [34] and double multi-scale feature pyramid network in [35]) can be used to automatically label RS images in an offline mode. In fact, promising automatic or semi-automatic annotation results have been reported in many recent studies. For example, in order to reduce the cost of labeling RS image, Yao *et al.* [36] proposed a weakly supervised model with an efficient high-level semantic feature transferring scheme. Zhuo *et al.* [37] automatically generated image annotations by label propagation based on a Bayesian-CRF model. To ensure their correctness, these automatically annotated images also should be pushed to volunteers for review. Overall, object detection methods or specialized automatic annotation methods can provide a useful way to improve the labeling efficiency of the tool.

V. CONCLUSION AND FUTURE WORK

This paper presents an open access web-based tool V-RSIR and it allows volunteers with minimal training in RS image identification to easily label, crop, edit and share RS images for RSIR. Besides, it also supports volunteers to view quantity and spatial distribution of new datasets, and as well as provides image review function with professional inspectors to ensure image quality. To evaluate the usability of the tool, 32 volunteers and 6 image inspectors are organized to label images from the Google image map by the tool. Finally, a new benchmark dataset VGoogle-RSIR covering 38 classes with least 1500 images per class is constructed. The evaluation results on five handcrafted low-level feature methods and four

deep learning high-level feature methods are consistent with our theoretical understanding and experimental results on the PatternNet dataset. This demonstrates the effectiveness and applicability of the V-RSIR tool.

A longer-term goal of V-RSIR tool is to simultaneously offer RS image single-label and multi-label function, and build a really large-scale benchmark dataset for RSIR as good as ImageNet. Therefore, our future work will concentrate on improving the functions of multi-label and automation.

ACKNOWLEDGMENT

The authors would like to thank the PatternNet dataset for its classification scheme. The authors would also like to thank the anonymous reviewers for their constructive and valuable comments that significantly contributed to improving this paper.

APPENDIX

See Table 7.

REFERENCES

- [1] B. Chaudhuri, B. Demir, L. Bruzzone, and S. Chaudhuri, "Region-based retrieval of remote sensing images using an unsupervised graph-theoretic approach," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 7, pp. 987–991, Jul. 2016.
- [2] X. Tang, X. Zhang, F. Liu, and L. Jiao, "Unsupervised deep feature learning for remote sensing image retrieval," *Remote Sens.*, vol. 10, no. 8, p. 1243, Aug. 2018.
- [3] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.
- [4] F. Ye, H. Xiao, X. Zhao, M. Dong, W. Luo, and W. Min, "Remote sensing image retrieval using convolutional neural network features and weighted distance," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1535–1539, Oct. 2018.
- [5] Z. Shao, K. Yang, and W. Zhou, "Performance evaluation of single-label and multi-label remote sensing image retrieval using a dense labeling dataset," *Remote Sens.*, vol. 10, no. 6, p. 964, 2018.
- [6] W. Zhou, S. Newsam, C. Li, and Z. Shao, "PatternNet: A benchmark dataset for performance evaluation of remote sensing image retrieval," *ISPRS-J. Photogramm. Remote Sens.*, vol. 145, pp. 197–209, Nov. 2018.
- [7] W. Xiong, Y. Lv, Y. Cui, X. Zhang, and X. Gu, "A Discriminative feature learning approach for remote sensing image retrieval," *Remote Sens.*, vol. 11, no. 3, p. 281, Feb. 2019.
- [8] Q. Bao and P. Guo, "Comparative studies on similarity measures for remote sensing image retrieval," in *Proc. IEEE Int. Conf. Syst., Man Cybern.*, Oct. 2004, pp. 1112–1116.
- [9] M. Ferecatu and N. Boujemaa, "Interactive remote-sensing image retrieval using active relevance feedback," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 4, pp. 818–826, Apr. 2007.
- [10] E. Aptoula, "Remote sensing image retrieval with global morphological texture descriptors," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 3023–3034, May 2014.
- [11] Y. Li, Y. Zhang, X. Huang, H. Zhu, and J. Ma, "Large-scale remote sensing image retrieval by deep hashing neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 950–965, Feb. 2018.
- [12] W. Zhou, S. Newsam, C. Li, and Z. Shao, "Learning low dimensional convolutional neural networks for high-resolution remote sensing image retrieval," *Remote Sens.*, vol. 9, no. 5, p. 489, 2017.
- [13] Y. Yang and S. Newsam, "Bag-of-visual-words and spatial extensions for land-use classification," in *Proc. 18th SIGSPATIAL Int. Conf. Adv. Geograp. Inf. Sys.*, San Jose, CA, USA, Nov. 2010, pp. 270–279.
- [14] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei, "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.

- [15] G. Sheng, W. Yang, T. Xu, and H. Sun, "High-resolution satellite scene classification using a sparse coding based multiple feature combination," *Int. J. Remote Sens.*, vol. 33, no. 8, pp. 2395–2412, 2012.
- [16] Q. Zou, L. Ni, T. Zhang, and Q. Wang, "Deep learning based feature selection for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 11, pp. 2321–2325, Nov. 2015.
- [17] B. Zhao, Y. Zhong, G.-S. Xia, and L. Zhang, "Dirichlet-derived multiple topic scene classification model for high spatial resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2108–2123, Apr. 2016.
- [18] G.-S. Xia, J. Hu, F. Hu, B. Shi, X. Bai, Y. Zhong, L. Zhang, and X. Lu, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.
- [19] H. Li, T. Chao, Z. Wu, C. Jie, and D. Min, "RSI-CB: A large scale remote sensing image classification benchmark via crowdsourced data," 2017, *arXiv:1705.10450*. [Online]. Available: <https://arxiv.org/abs/1705.10450>
- [20] T. Liu and X. Chen, "Application of deep learning in globeland30-2010 product refinement," *Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci.*, vol. 3, pp. 1111–1116, Apr. 2018.
- [21] J. Chen, J. Chen, A. Liao, X. Cao, L. Chen, X. Chen, C. He, G. Han, S. Peng, M. Lu, and W. Zhang, "Global land cover mapping at 30 m resolution: A POK-based operational approach," *ISPRS J. Photogramm. Remote Sens.*, vol. 103, pp. 7–27, May 2015.
- [22] B. C. Russell, A. Torralba, K. P. Murphy, and W. T. Freeman, "LabelMe: A database and Web-based tool for image annotation," *Int. J. Comput. Vis.*, vol. 77, nos. 1–3, pp. 157–173, 2008.
- [23] M. Haklay and P. Weber, "OpenStreetMap: User-generated street maps," *IEEE Pervasive Comput.*, vol. 7, no. 4, pp. 12–18, Oct. 2008.
- [24] S. Fritz et al., "Geo-Wiki.Org: The use of crowdsourcing to improve global land cover," *Remote Sens.*, vol. 1, no. 3, pp. 345–354, 2009.
- [25] G. S. Xia, X. Y. Tong, F. Hu, Y. Zhong, M. Datcu, and L. Zhang, "Exploiting deep features for remote sensing image retrieval: A systematic investigation," 2017, *arXiv:1707.07321*. [Online]. Available: <https://arxiv.org/abs/1707.07321>
- [26] L. Bastin, G. Buchanan, A. Beresford, J. F. Pekel, and G. Dubois, "Open-source mapping and services for Web-based land-cover validation," *Ecol. Inform.*, vol. 14, pp. 9–16, Mar. 2013.
- [27] L. Kumar and O. Mutanga, "Google earth engine applications since inception: Usage, trends, and potential," *Remote Sens.*, vol. 10, no. 10, p. 1509, Sep. 2018.
- [28] H. Li, W. Wan, Y. Fang, S. Zhu, X. Chen, B. Liu, and Y. Hong, "A google earth engine-enabled software for efficiently generating high-quality user-ready Landsat mosaic images," *Environ. Model. Softw.*, vol. 112, pp. 16–22, Feb. 2019.
- [29] B. Chaudhuri, B. Demir, S. Chaudhuri, and L. Bruzzone, "Multilabel remote sensing image retrieval using a semisupervised graph-theoretic method," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 1144–1158, Feb. 2018.
- [30] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2015, pp. 3431–3440.
- [31] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask r-cnn," in *Proc. IEEE Int. Conf. Comput. Vis.*, Jun. 2017, pp. 2961–2969.
- [32] T. Zhang, L. Guo, K. Li, C. Jing, Y. Yin, D. Zhu, G. Cui, L. Li, and T. Liu, "Predicting functional cortical ROIs via DTI-derived fiber shape models," *Cerebral Cortex*, vol. 22, no. 4, pp. 854–864, 2011.
- [33] J. Han, X. Ji, X. Hu, D. Zhu, K. Li, X. Jiang, G. Cui, L. Guo, and T. Liu, "Representing and retrieving video shots in human-centric brain imaging space," *IEEE Trans. Image Process.*, vol. 22, no. 7, pp. 2723–2736, Jul. 2013.
- [34] G. Cheng, P. Zhou, and J. Han, "Learning rotation-invariant convolutional neural networks for object detection in VHR optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 12, pp. 7405–7415, Dec. 2016.
- [35] X. Zhang, K. Zhu, G. Chen, X. Tan, L. Zhang, F. Dai, P. Liao, and Y. Gong, "Geospatial object detection on high resolution remote sensing imagery based on double multi-scale feature pyramid network," *Remote Sens.*, vol. 11, no. 7, p. 755, Mar. 2019.
- [36] X. Yao, J. Han, G. Cheng, X. Qian, and L. Guo, "Semantic annotation of high-resolution Satellite images via weakly supervised learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3660–3671, Jun. 2016.

- [37] X. Zhuo, F. Fraundorfer, F. Kurz, and P. Reinartz, "Automatic annotation of airborne images by label propagation based on a Bayesian-CRF model," *Remote Sens.*, vol. 11, no. 2, p. 145, Jan. 2019.



DONGYANG HOU was born in 1986. He received the Ph.D. degree from the China University of Mining and Technology, China, in 2016. From 2016 to 2019, he was a Lecturer with the College of Geography and Environment, Shandong Normal University, China. He is currently a Lecturer with the School of Geosciences and Info-Physics, Central South University, China. His main research interests include spatial-temporal data crawling, mining, GIS software development, and remote sensing image retrieval. He serves as a Secretary for the International Society for Photogrammetry and Remote Sensing (ISPRS) Inter Commission Working Group IV/III.



ZELANG MIAO (S'15–M'16) received the B.S. degree in surveying engineering and the M.S. degree in geodesy and surveying engineering from the China University of Mining and Technology, Xuzhou, China, in 2009 and 2014, respectively, and the Ph.D. degree in satellite image processing from The Hong Kong Polytechnic University, Hong Kong, in 2016.

He was a Visiting Ph.D. Student with the Department of Industrial and Information Engineering, University of Pavia, Pavia, Italy, in 2015. He is currently an Associate Professor with the School of Geosciences and Info-Physics, Central South University, Changsha, China. He has authored or coauthored more than 30 papers on international and peer-reviewed scientific journals. His research interests include pattern recognition, land cover and land use mapping, and global/regional urbanization.

Dr. Miao serves as a Referee for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, the IEEE JOURNAL OF SELECTED TOPICS IN REMOTE SENSING APPLICATIONS, and *Remote Sensing*.



HUAQIAO XING was born in Shandong, in 1988. He received the M.S. degree in GIS from the Beijing University of Architecture, in 2012, and the Ph.D. degree in GIS from the China University of Mining and Technology, Beijing, in 2017. He joined Shandong Jianzhu University. His current research interests include land cover change detection, geo-service computing, and global discrete grid modeling.



HAO WU was born in 1983. He received the Ph.D. degree from Wuhan University, China, in 2012. He is currently a Senior Engineer with the National Geomatics Center of China, China. He has been serving as a Chair for the ISPRS Inter Commission Working Group IV/III. His current research interests include global mapping updating, verification, and interoperability.

...